

Gearbox Inventory & Purchasing Study

Reconciling two China purchase invoices against the customer order book — what to buy more of, what to stop buying, and whether stock is sufficient.

SUPPLIER: HANGZHOU FIXEDSTAR BUYER: I-MAK / IRAS PERIOD: DEC 2025 – JUL 2026 PREPARED: 11 AUG 2026

BOTTOM LINE

Stock is **not** insufficient — it is roughly **8× oversized**. You have **12,900 gearboxes** on order from China against **1,623 customer order lines**. About **\$420,612** — 86% of the gearbox value — is surplus beyond recorded demand. Genuine shortages exist, but they total only **143 pieces / ≈\$5,040**. The action is not to buy more; it is to stop, rebalance, and buy nine small flange combinations.

GEARBOXES ON ORDER 12,900 pcs across 203 distinct SKUs	CUSTOMER ORDER LINES 1,623 from SP_SIPARISLER	OVERALL COVER 7.95× supply ÷ demand
SURPLUS VALUE \$420,612 86% of gearbox value	TRULY SHORT \$5,040 143 pcs, 9 combinations	TOTAL PURCHASE VALUE \$495,100 after removing 1 duplicate order

01 What the three files actually contain

The two *Fixedstara* files are commercial invoices from Hangzhou Fixedstar to I-MAK. File 1 is invoice **FS2606043L** dated 9 Jun 2026, totalling **\$238,991.25**. File 2 carries no invoice number, is dated 17 Jun 2026 and totals **\$257,628.46**. Every line in both files was parsed and re-summed; both reproduce the printed invoice totals exactly, so the extraction is verified.

They are **not** duplicates of each other — they cover different order references, with one exception. Order **FCM260701** (four input-flange lines, 400 pcs, \$1,520) appears identically in both files and has been counted once. Orders **WX4847** and **WX5008** are split across the two files, each carrying different lines of the same order — those are genuine complements, not duplicates.

The third file, **SP_SIPARISLER**, is the customer order book: 1,623 rows, one order line each. It contains only three columns — model, ratio, IEC flange — and the last two are mostly empty.

FILE	ORDER REF	WX	LINES	PIECES	VALUE USD	STATUS
1	FCM251226	WX4417	12	1,120	33,525.00	COUNTED

FILE	ORDER REF	WX	LINES	PIECES	VALUE USD	STATUS
1	ADD ON DEC.30	WX4422	8	99	5,313.50	COUNTED
1	ADD ON JAN.07	WX4437	47	1,135	89,139.30	COUNTED
1	ADD ON MAR.02	WX4734	117	1,083	43,002.75	COUNTED
1	ADD ON MAR.23	WX4847	47	2,616	60,625.70	SPLIT ACROSS FILES
1	FCM260508	WX5008	2	23	5,865.00	COUNTED
1	FCM260701	WX5252	4	400	1,520.00	DUPLICATE — EXCLUDED
2	FCM251226	WX4847	49	2,446	47,437.30	COUNTED
2	ADD ON APR.14	WX4909	1	500	35,500.00	COUNTED
2	FCM260508	WX5008	82	1,747	63,036.30	COUNTED
2	FCM260601	WX5112	43	1,759	35,983.10	COUNTED
2	FCM260603	WX5123	27	680	17,117.50	COUNTED
2	FCM260616	WX5174	4	650	16,291.00	COUNTED
2	FCM260617	WX5183	135	977	35,867.26	INTERNAL OVERLAP — VERIFY
2	FCM260701 + SPECIAL	WX5204	5	440	6,396.00	COUNTED
Total counted			579	15,275	495,099.71	

02 Method and the one assumption that matters

The customer file has **no quantity column**, no dates, no prices and no customer names. Each of the 1,623 rows is treated as demand for **1 piece**. This is the single assumption the whole study rests on, so here is what happens if it is wrong:

IF EACH ORDER LINE =	TOTAL DEMAND (PCS)	COVER RATIO	READING
1	1,623	7.95×	EXTREME OVERSTOCK
2	3,246	3.97×	HEAVY OVERSTOCK
3	4,869	2.65×	OVERSTOCK
5	8,115	1.59×	COMFORTABLE
7.95	12,903	1.00×	EXACTLY BALANCED
10	16,230	0.79×	SHORTAGE

Break-even is **7.95 pieces per order line**. For the current purchasing to be justified, the average customer order line would have to be almost eight units. If you can export the quantity column from your ERP, that one field converts this study from directional to exact — but even at 3 pcs/line the portfolio is still 2.65× overbought.

SECOND LIMITATION

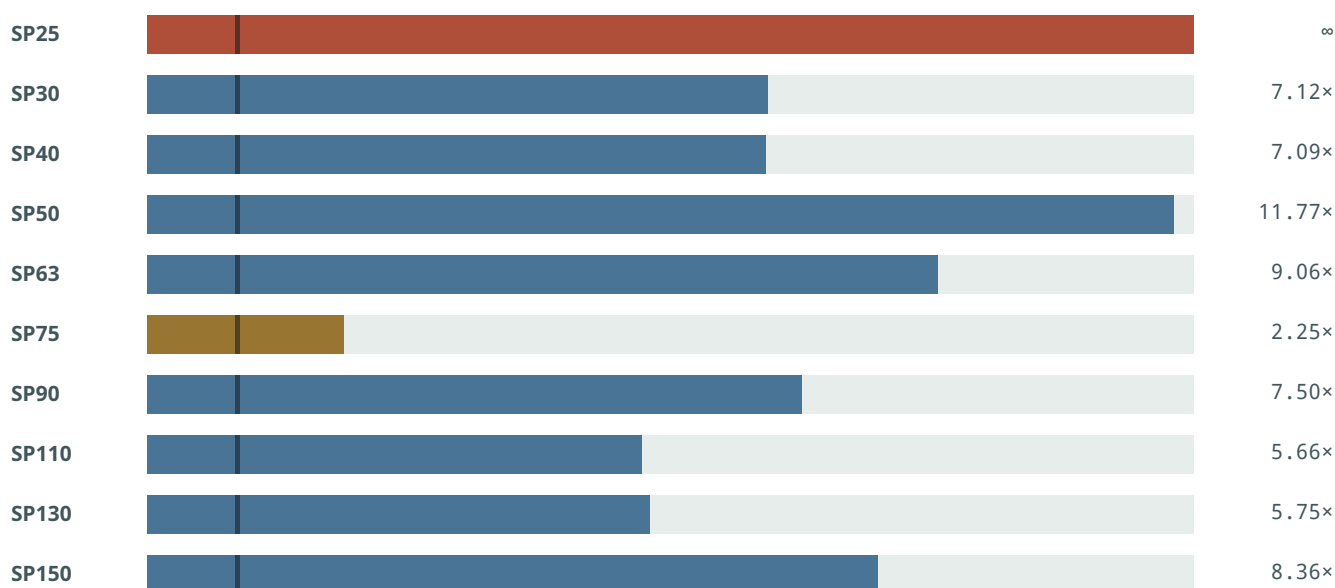
Ratio is filled in only **134 of 1,623** customer rows — and every one of them says 7.5. IEC flange is filled in only **275 of 1,623** rows. Size-level conclusions are therefore solid; ratio- and flange-level conclusions are projections from those samples and are labelled as such.

03 Size-level result — supply against demand

Every single model is overbought. **SP50** is the worst offender in absolute terms — 4,474 pieces against 380 order lines, \$106,580 of surplus. **SP75** is the tightest at 2.25× and is the one model where demand is genuinely close to supply. **SP25** has 59 pieces on order and **zero** recorded customer demand.

MODEL	ON ORDER	DEMAND	COVER	SURPLUS	UNIT FOB	SURPLUS VALUE	VERDICT
SP25	59	0	∞	59	\$14.90	\$879	NO DEMAND
SP30	1,103	155	7.12×	948	\$15.04	\$14,260	OVERSTOCK
SP40	1,807	255	7.09×	1,552	\$17.34	\$26,912	OVERSTOCK
SP50	4,474	380	11.77×	4,094	\$26.03	\$106,580	HEAVY OVERSTOCK
SP63	3,106	343	9.06×	2,763	\$35.60	\$98,371	OVERSTOCK
SP75	485	216	2.25×	269	\$52.00	\$13,988	WATCH
SP90	1,147	153	7.50×	994	\$71.00	\$70,574	OVERSTOCK
SP110	334	59	5.66×	275	\$115.83	\$31,852	OVERSTOCK
SP130	293	51	5.75×	242	\$151.00	\$36,542	OVERSTOCK
SP150	92	11	8.36×	81	\$255.00	\$20,655	OVERSTOCK
Σ	12,900	1,623	7.95×	11,277		\$420,612	

COVER RATIO BY MODEL (VERTICAL LINE = 1.0× BALANCE)



Bars capped at 12×. SP75 is the only model under 3× cover; SP50 and SP63 alone hold 58.8% of all pieces and \$326k of value.

04 Portfolio mix — buying the wrong proportions

Separate from the overall volume problem, the *shape* of the purchasing does not match the shape of demand. The table compares each model's share of purchases with its share of customer orders.

MODEL	SHARE OF DEMAND	SHARE OF PURCHASES	GAP	PIECES MISALLOCATED
SP25	0.0%	0.5%	+0.5	+59
SP30	9.6%	8.6%	-1.0	-129
SP40	15.7%	14.0%	-1.7	-220
SP50	23.4%	34.7%	+11.3	+1,454
SP63	21.1%	24.1%	+3.0	+380
SP75	13.3%	3.8%	-9.5	-1,232
SP90	9.4%	8.9%	-0.5	-69
SP110	3.6%	2.6%	-1.0	-135
SP130	3.1%	2.3%	-0.8	-112
SP150	0.7%	0.7%	+0.0	+5

Two numbers drive everything: **SP50 is over-weighted by 11.3 points** ($\approx 1,454$ pieces too many) and **SP75 is under-weighted by 9.5 points** ($\approx 1,232$ pieces too few relative to its demand share). Shifting future orders from SP50 to SP75 fixes the mix without increasing total spend.

Where the volume went, by ratio

RATIO I	PIECES	SHARE
i = 100	3,289	25.5%
i = 50	1,795	13.9%
i = 40	1,553	12.0%
i = 15	1,306	10.1%
i = 20	1,067	8.3%
i = 30	926	7.2%
i = 25	719	5.6%
i = 7.5	698	5.4%
i = 80	504	3.9%
i = 10	384	3.0%
i = 60	325	2.5%
i = 5	294	2.3%

A quarter of everything bought — **3,289 pieces at i=100** — sits in the slowest ratio. The customer file gives no ratio evidence to support that concentration.

05 What to buy more of

Projecting the 275 customer rows that do carry an IEC flange onto each model's total demand gives the shortage list below. It is short and it is cheap: **143 pieces, about \$5,040**.

MODEL	FLANGE (PAM)	PROJECTED NEED	ON ORDER	COVER	BUY	COST USD
SP90	112B14	26	8	0.31x	18	\$1,278

MODEL	FLANGE (PAM)	PROJECTED NEED	ON ORDER	COVER	BUY	COST USD
SP90	80B14	26	5	0.20×	21	\$1,491
SP30	80B14	43	0	0.00×	43	\$647
SP50	90B14	27	0	0.00×	27	\$703
SP30	90B14	9	0	0.00×	9	\$135
SP40	90B14	8	0	0.00×	8	\$139
SP75	71B14	7	0	0.00×	7	\$364
SP63	63B14	6	0	0.00×	6	\$214
SP40	80B14	4	0	0.00×	4	\$69
SP63	71B14	66	87	1.31×	—	—
SP75	100B14	37	50	1.34×	—	—
SP75	90B14	97	157	1.62×	—	—
Σ					143	\$5,040

The nine zero-or-near-zero rows are real gaps: SP90 with 80B14 and 112B14 flanges, SP30 with 80B14 and 90B14, SP50 with 90B14, SP40 with 80B14 and 90B14, SP63 with 63B14, SP75 with 71B14. None of these were ordered in meaningful volume.

Ratio 7.5 — the one ratio the customer file confirms

134 customer rows explicitly request i=7.5. Against them: **SP130 has zero pieces at i=7.5** (3 requested), SP75 has 6 against 13 requested (0.46×), and SP90 has 5 against 7 (0.71×). Add roughly **12 pieces** here.

MODEL	REQUESTS AT I=7.5	ON ORDER AT I=7.5	COVER
SP30	23	125	5.43×
SP40	22	85	3.86×
SP50	36	276	7.67×
SP63	27	166	6.15×
SP75	13	6	0.46×
SP90	7	5	0.71×
SP110	3	10	3.33×
SP130	3	0	0.00×

06 What to stop buying

These ten combinations carry cover between 4.5× and 25.6×. **SP50 with 71B14 alone holds 3,127 pieces against a projected need of 122** — twenty-five years of demand at the current rate.

MODEL	FLANGE (PAM)	PROJECTED NEED	ON ORDER	COVER
SP50	71B14	122	3,127	25.6×
SP50	63B14	9	190	21.0×
SP90	100B14	42	811	19.1×

MODEL	FLANGE (PAM)	PROJECTED NEED	ON ORDER	COVER
SP63	80B14	138	1,888	13.7×
SP40	63B14	61	514	8.5×
SP63	90B14	133	939	7.1×
SP40	71B14	182	1,283	7.0×
SP30	63B14	103	712	6.9×
SP130	112B14	51	276	5.4×
SP50	80B14	222	1,000	4.5×

STOP ENTIRELY

- **All B5 flanges** — 706 pcs, \$51,006. Not one of the 275 customer rows with flange data asks for B5; every single one is B14.
- **SP25** — 59 pcs, \$879. Zero recorded demand.
- **SP50 / SP63 new orders** — 7,580 pcs already on order, \$326k, 58.8% of the whole portfolio.

TRIM HARD

- **Ratio i=100** — 3,289 pcs, a quarter of all stock, in the slowest ratio with no demand evidence.
- **SP50 i=100 / 71B14** — 1,946 pcs, \$50,596. The largest single SKU in the book, against total SP50 demand of 380 lines.
- **Micro-lots** — 45 SKUs of 5 pieces or fewer: 174 pcs but \$12,458 tied up and 45 separate stock locations to manage.

B5: SP63 90B5 135 · SP50 80B5 92 · SP75 90B5 86 · SP150 132B5 83 · SP90 100B5 81 · SP50 71B5 65 · SP30 63B5 41 · SP90 90B5 35 · SP63 80B5 30

07 Data quality flags to resolve

- Order **FCM260701** is present identically in both files (400 pcs, \$1,520). Counted once here — confirm it was invoiced once.
- Inside file 2, blocks **WX5008** and **WX5183** share 45 identical (model / ratio / flange / quantity) lines totalling **287 pieces**. The invoice total adds both. Verify this is a genuine repeat order and not a double entry.
- File 2's packing list sheet itemises only **60 rows** (WX4847 and WX4909) but its footer states 65,795 kg gross, 94.84 CBM and **124 pallets**. The itemised detail does not cover the full shipment.
- File 1, line 43 reads *QTY:195* in its description but is invoiced at **162 pieces**. The invoiced figure was used.
- File 2's invoice sheet has **no invoice number**, while its own packing list quotes FS2606043L — the same number as file 1.
- The customer file carries no quantity, date, price or customer column. Adding quantity and date is the highest-value fix available.

08 Action plan

01	Freeze SP50 and SP63 Place no new orders on these two models until stock falls below 2× cover. This alone protects \$326,000 of working capital.
02	Cancel or defer what is still open Any unshipped portion of the i=100 and B5 lines — up to 3,995 pieces and roughly \$101,000 combined — is the cheapest saving available.
03	Place one small corrective order 143 pieces across nine flange combinations, plus ~12 pieces at i=7.5. Total ≈\$5,040. This is the entire genuine shortage.
04	Rebalance toward SP75 SP75 is 13.3% of demand but 3.8% of purchases and the only model under 3× cover. Move future SP50 budget here.
05	Fix the customer data Export quantity and order date with the order book. Cover ratios become exact instead of directional, and monthly run-rate becomes computable.
06	Reconcile the two flagged overlaps Confirm FCM260701 and the WX5008/WX5183 repeat with the supplier before the next payment run.

Sources: Fixedstara Geçilen Siparişler 1 & 2 (commercial invoices, sheet inv; file 2 also contains sheet p1) and SP_SIPARISLER (sheet REPORTLISTMAT). All 583 invoice lines were parsed programmatically; the recomputed totals match the printed invoice totals of \$238,991.25 and \$257,628.46 to the cent. Demand figures assume one piece per customer order line — see section 2. Values are FOB Shanghai in USD.